

HEALTHCONNECT — WEEK 8 FINAL DATA ANALYTICS REPORT

By Ajibola Ayobami (Data Analysis Intern)

Final Analytics, Dashboard & Business Insights

1. Executive Summary

The HealthConnect Data Analytics project analysed 5,000 appointment records to understand appointment attendance, no-show behaviour and factors associated with missed appointments.

The final analysis identified an overall **48.46% no-show rate**, with **46.3% of appointments attended** and **5.3% cancelled**.

The strongest observed analytical pattern was the relationship between **previous no-show history and current no-show behaviour**. Appointments involving patients with previous no-shows generally showed higher current no-show rates.

Other observed patterns were identified across reminder status, appointment type, age group and booking lead time. These findings were validated during Week 7 against the underlying dataset and incorporated into the final Power BI dashboard.

The final analytics output provides HealthConnect with a decision-support view that can help identify appointment segments requiring closer monitoring, strengthen appointment-management processes and support evidence-based operational decisions.

2. Project Objective

The objective of the Data Analytics component was to analyse HealthConnect appointment data and provide reliable insights that could support efforts to improve appointment attendance and healthcare support.

The final output focused on:

- Understanding appointment outcomes.
- Measuring no-show and attendance rates.
- Identifying patterns associated with missed appointments.
- Validating important analytical findings.
- Presenting findings through an interactive Power BI dashboard.

- Translating findings into practical business recommendations.
-

3. Final Dataset

The analysis used the HealthConnect appointment dataset containing:

5,000 appointment records

The analysis considered variables including:

- Appointment outcome
- Previous appointments
- Previous no-shows
- Reminder status
- Appointment type
- Age
- Age group
- Booking lead time
- Appointment day
- Appointment time
- Distance to clinic
- Waiting time
- Gender

The same HealthConnect dataset was maintained throughout the project rather than introducing a new dataset during the final stage, consistent with the Week 8 requirement to build on previous work.

4. Final KPI Results

KPI	Final Result
Total Appointments	5,000
No-Show Count	2,423
No-Show Rate	48.46%
Attendance Rate	46.3%
Cancellation Rate	5.3%

These KPIs were validated during Week 7 and retained in the final dashboard.

5. Final Analytical Findings

5.1 Previous No-Show History

Previous no-show history was the clearest observed indicator associated with current no-show behaviour.

Previous No-Shows	No-Show Rate
0	43.51%
1	53.49%
2	59.36%
3	67.95%
4	66.67%
5	100.00%

The pattern generally increases from 0 to 3 previous no-shows.

However, the groups with four and five previous no-shows contain very few appointments, so those percentages should **not** be treated as stable population estimates.

Business interpretation

Previous no-show history can be used as an important indicator for identifying appointments that may require closer monitoring and additional engagement.

5.2 Reminder Status

The final dashboard showed:

- **No reminder:** 51.39% no-show rate
- **Reminder sent:** 47.36% no-show rate

Appointments with a recorded reminder therefore had a lower observed no-show rate.

However, this is an **association, not evidence that reminders caused the difference.**

Business interpretation

HealthConnect should continue monitoring reminder coverage and appointment outcomes while using further testing before making causal claims about reminder effectiveness.

5.3 Appointment Type

Observed no-show rates differed across appointment types:

- Follow-up: **51.23%**
- Diagnostic Test: **49.75%**
- Specialist Consultation: **47.44%**
- General Consultation: **46.54%**

Business interpretation

Follow-up appointments showed the highest observed no-show rate among the appointment categories analysed and therefore provide an area for continued monitoring and process review.

5.4 Age Group

No-show rates also varied across age groups.

The final dashboard showed the highest observed rates among some younger and middle-aged groups, while the 65+ group showed a lower observed rate.

These differences are descriptive and should be used for monitoring rather than interpreted as evidence that age itself causes missed appointments.

5.5 Booking Lead Time

The final analysis showed a broad upward pattern in observed no-show rates as booking lead time increased.

Business interpretation

Longer booking lead times may be useful as a monitoring indicator, particularly when considering appointment confirmation closer to the scheduled date.

5.6 Distance and Waiting Time

Distance to the clinic and waiting time were included as operational factors.

The final dashboard presents average distance and waiting time by appointment outcome.

These variables remain useful for monitoring patient appointment experience, but the analysis does not establish that distance or waiting time directly causes no-shows.

6. Final Dashboard

The final Power BI dashboard provides:

- Total appointment KPI
- No-show rate
- Attendance rate
- Cancellation rate
- No-show rate by reminder status

- No-show rate by previous no-show history
- No-show rate by appointment type
- No-show rate by age group
- Average distance and waiting time
- No-show rate by booking lead time
- Appointment Type filter
- Appointment Outcome filter
- Gender filter
- Age Group filter
- Key decision-support findings
- Recommended actions
- Previous no-show validation table
- Small-sample caution note

The dashboard therefore combines **KPI monitoring, analytical exploration and decision support** in one final output.

7. Final Business Recommendations

1. Prioritise patients with previous no-show history

HealthConnect should consider additional appointment confirmation and engagement for patients with a history of missed appointments.

2. Strengthen reminder processes

Continue monitoring reminder coverage and observed appointment outcomes, while recognising that the current analysis does not establish causality.

3. Monitor appointment types

Follow-up appointments showed the highest observed no-show rate among the appointment categories analysed and should receive continued monitoring.

4. Monitor longer booking lead times

HealthConnect could consider additional confirmation closer to the appointment date when appointments are booked further in advance.

5. Use patient segmentation

Age group and other appointment characteristics can support monitoring and targeted operational attention.

6. Monitor operational factors

Distance and waiting time should continue to be monitored as part of the wider patient appointment experience.

7. Use evidence carefully

Very small groups should not drive major decisions, and observed associations should not automatically be treated as causal relationships.

8. Business Value

The Data Analytics contribution provides HealthConnect with a clearer evidence base for understanding missed appointments.

Instead of looking only at the overall no-show percentage, the final dashboard allows decision-makers to examine no-show behaviour across different appointment and patient characteristics.

This can help HealthConnect:

- Identify appointment segments requiring closer monitoring.
 - Support targeted appointment-management strategies.
 - Monitor reminder and attendance patterns.
 - Identify potential operational areas for improvement.
 - Track key appointment KPIs.
 - Support evidence-based decision-making.
-

9. Validation and Quality Assurance

The final analytics output was developed through multiple stages:

Week 5: Exploratory analysis and dashboard development

↓

Week 6: Advanced analysis and deeper segmentation

↓

Week 7: Testing, KPI validation and dashboard refinement

↓

Week 8: Final analytics and decision support

During Week 7, important dashboard figures were checked against the underlying dataset.

The previous no-show analysis was also validated using a detailed table showing:

- Previous no-shows
- Total appointments
- No-show count
- No-show rate

This provided additional evidence supporting the final analytical conclusion.

10. Limitations

The following limitations should be considered when interpreting the final analysis:

1. The dataset is observational, so relationships identified in the analysis should not be interpreted as causal effects.
2. The groups with four and five previous no-shows contain very few observations.
3. Distance and waiting-time variables contain limited missing values.
4. The analysis identifies patterns and associations but does not independently establish the underlying causes of missed appointments.
5. Further testing would be appropriate before using these findings for automated or high-impact decision-making.
6. Cross-track feedback was not received during the project period, so no cross-track testing outcome is claimed.

11. Cross-Track Contribution

The Data Analytics outputs are relevant to other HealthConnect workstreams, particularly Data Science and Project Management.

Potentially useful analytical outputs include:

- Previous no-show history
- Reminder status
- Appointment type
- Age group
- Booking lead time
- Validated KPI results

These findings can provide supporting information for future modelling, project decisions and solution development.

However, **no completed cross-track testing result is claimed**, because feedback from another track was not received during the submission period.

This outstanding dependency should be followed up during the next stage.

The Week 8 guidance specifically expects relevant validated findings to be provided to other tracks and integrated where applicable.

12. Final Deliverables

The completed Data Analytics work consists of:

-  HealthConnect Power BI dashboard
-  Final KPI calculations
-  Validated analytical findings
-  Previous no-show validation table
-  Business insights
-  Business recommendations
-  Dashboard filters

-  Testing and refinement evidence
 -  Limitations
 -  Executive summary
 -  Final analytics report
-

13. Final Conclusion

The HealthConnect Data Analytics project progressed from exploratory analysis to advanced analytical investigation, testing and final decision support.

The final analysis identified a **48.46% observed no-show rate** across 5,000 appointments. Previous no-show history emerged as the clearest observed indicator associated with current no-show behaviour, while reminder status, appointment type, age group and booking lead time also showed useful patterns for monitoring.

The final Power BI dashboard brings these findings together into an interactive decision-support tool.

The analysis provides HealthConnect with a stronger evidence base for identifying appointment-management priorities while maintaining appropriate caution around causality, small samples and remaining data limitations.